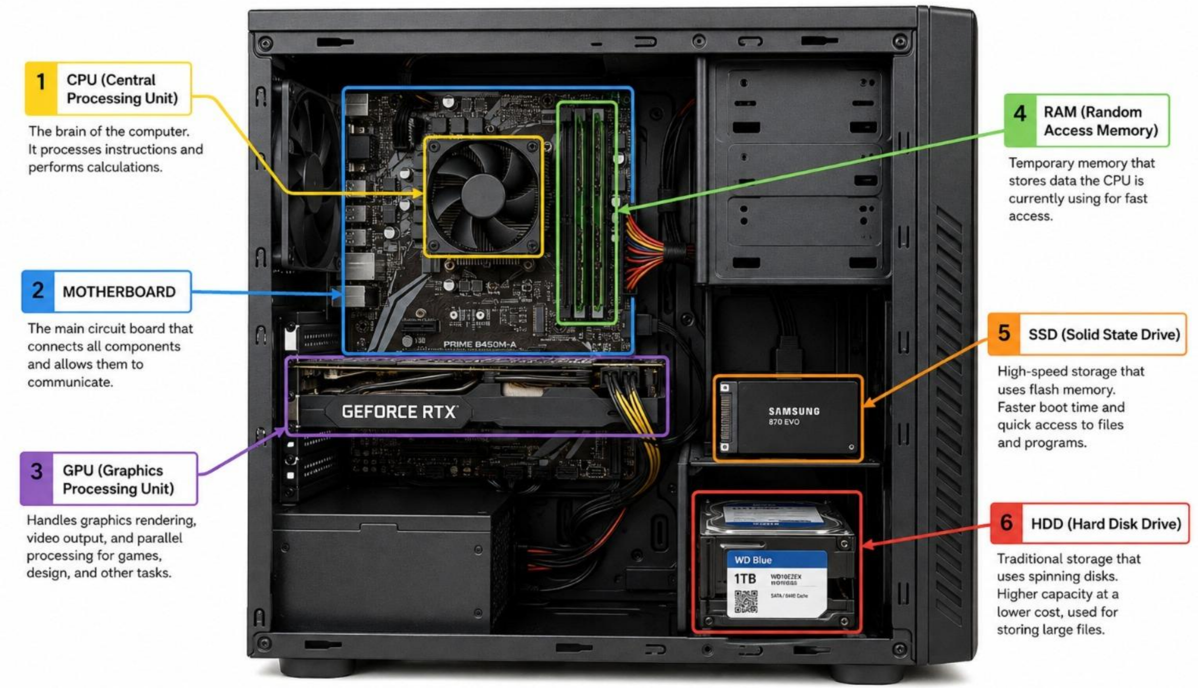
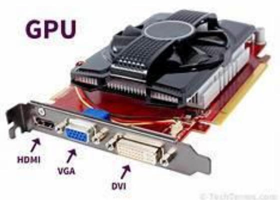
Assignment - 1.1

Day 1 : Overview of hardware components Theory : Introduction to motherboards, CPUs, RAM, GPUs, storage (HDD, SSD). Importance of each component in a computer system. Assignment: Create a diagram of a PC with labeled hardware components.

Task - 1

Draw a labelled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.





Task – 2

Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Features	HDD	SSD
Speed	Slower	Faster
Durability	Less Shock-resistant	More Shock-resistant
Use Cases	Large storage, Backups	Gaming, laptops, OS and Applications

- **HDD (Hard Disk Drive)** : HDD is a storage device used to save files, photos, videos, and other data. It has moving parts inside, so it is **slower** and can be damaged by physical shock. HDD is mainly used for **large storage and backups**. Since HDD has moving parts, it takes more time to access data and is therefore slower.

- **SSD (Solid State Drive) :** SSD is a storage device that uses memory chips to save data. It has no moving parts, so it is faster and more durable than HDD. SSD is commonly used in laptops, gaming PCs, and for operating systems and applications. SSD is an electronic storage device based on **flash memory**.

Task – 3

Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

- **CPU (Central Processing Unit) :-**
  CPU is called the **brain** of the computer. It performs calculations, processes game data, and handles instructions like player movement and game logic. A powerful CPU helps the game run smoothly and quickly.

  Reason :- If the CPU is weak, the game may become slow and the frame rate may drop because the computer cannot process the game calculations quickly.

- **RAM (Random Access Memory) :-**
  RAM is a **temporary memory** used by the computer while the game is running. It stores the game data that is currently needed, such as maps, characters, and other game information.

  Reason :- If there is not enough RAM, the game may lag or become slow because the computer does not have enough space to keep the required game data.

- **GPU (Graphics Processing Unit) :-**
  GPU is responsible for **processing and displaying** the graphics of the game. It creates characters, textures, lighting, maps, colours, and other visual effects. A powerful GPU provides better graphics and smoother gameplay.

  Reason :- If the GPU is weak, the game may look blurry or run slowly because it cannot process the graphics and visual effects properly.

Task - 4

Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Hardware	Reason for Upgrading
GPU	The GPU is responsible for creating and displaying the game's graphics, such as players, stadiums, lighting, and animations. If the GPU is powerful, FIFA will run with better graphics and smoother frame rates. If the GPU is weak, the game may lag or the graphics may look blurry.
CPU	The CPU processes the game's instructions, calculations, player movements, and game logic. If the CPU is powerful, the game can process these tasks quickly. If the CPU is weak, the game may become slow and the frame rate may drop.
RAM	RAM stores the game data temporarily while FIFA is running. More RAM helps the computer handle the game and other background programs smoothly. If there is not enough RAM, the game may lag or freeze because the computer does not have enough memory for the required data.
SSD	A SSD stores the game and helps the computer load the game and its files faster. If I use an SSD instead of a slower HDD, FIFA will load faster and the overall system will respond more quickly.